



Environmental Hypothermia

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by the HandTevy.
- Pediatric hypotension is defined as SBP < 70 + (age in years x 2) mmHg

Signs & Symptoms:

- Cold, clammy
- Shivering
- Mental status changes
- Extremity pain or sensory abnormality
- Bradycardia
- Hypotension or shock

Differential:

- Metabolic disorder (hypoglycemia, hypothyroidism)
- Toxins
- Environmental exposure
- Shock
- Sepsis

Clinical Management Options

P	P	P	P	P	P	• Oxygen: target SpO₂: 92-94% • Prevent further heat loss: Remove wet clothing, move to a warm environment, blankets & heat packs as needed, allow active movement after 20min if >89.6°F(32°C) • Obtain core temperature, if able: ◦ Handle very gently if <88°F(<30°C) due to risk of arrhythmias • BGL assessment • In cardiac arrest with temperature <88°F(<30°C), limit defibrillations to one shock & one Epinephrine until core temp greater than 88°F (>30°C) and continue CPR ◦ Consider early transport for active rewarming
L	L	L	L	L	L	• Increase the temperature of transport compartment
1	2	3	4	5	6	• Vascular access • Warm IV Isotonic Crystalloid if available – Use of blood warmer if available

Consult Online Medical Control As Needed

Pearls:

<u>Stages of Hypothermia</u>	<u>Temperature</u>	<u>Description</u>
Mild	35-32C (95-89F)	Conscious, shivering
Moderate	32-28C (89-82F)	Impaired consciousness, no longer shivering
Severe	28-24C (82-75F)	Unconscious, no shivering, vital signs present
Profound	<24C (<75F)	Cardiac arrest or absence of vital signs

- During warming, cold blood may re-enter central circulation causing a subsequent decrease in body temperature.

CARDIAC ARREST

- In suspected cardiac arrest due to environmental hypothermia, apparent rigor mortis, or dependent lividity are not criteria for withholding resuscitation.
- Once resuscitation begins, these patients should be transported for advanced rewarming and are generally not candidates for prehospital TOR (unless core temperature > 32°C/90°F).
 - Prolonged CPR is not indicated in patients who are thought to have experienced cardiac arrest **before** cooling (e.g. Cardiac arrest with immediate submersion, witnessed normothermic arrest, etc.).
- Perform prolonged assessment (60 seconds) for pulse and respirations before confirming cardiac arrest in hypothermic patients



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Criteria for Withholding Resuscitation in Hypothermic Arrest

- Chest wall not compressible because body is frozen solid
- Injuries incompatible with life (e.g., decapitation)
- Patient exhibits signs of being frozen (e.g., ice formation in the airway)

Bradycardia

- Hypothermia may produce severe physiologic bradycardia. Do not treat unless profound hypotension unresponsive to fluids.
 - TCP and atropine are not likely to be effective; early active warming is required